

AI Multilingual Text Summarizer

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An AI-powered full-stack web application for summarizing text & PDFs with multilingual support, keyword extraction, and document Q&A.;

Tech Stack: React, Tailwind CSS, Node.js, Express, Transformers (T5, NLLB)

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About Project

Web-based AI tool to summarize long text and PDFs into concise insights.

Supports multiple Indian languages using translation models.

Provides keywords and chat-based document understanding.

Designed for productivity, research, and quick decision-making.

Problem Statement

Users struggle to read lengthy documents efficiently.

Manual summarization is time-consuming and inconsistent.

Language barriers reduce accessibility of information.

Need for intelligent automation to extract insights quickly.

Solution & Approach

Used T5 transformer model for high-quality summarization.

NLLB model enables multilingual translation support.

PDF parsing via pdfjs for text extraction.

Frontend built with modern responsive UI using Tailwind.

System Architecture

Layer	Technology
Frontend	React + Tailwind
Backend	Node.js + Express
AI Models	T5 + NLLB
File Handling	Multer + PDFJS

Key Features

Text & PDF summarization with adjustable length.

Multilingual output (Hindi, Tamil, Punjabi, etc.).

Keyword extraction for quick insights.

Chat with PDF for contextual Q&A.

Download summary as PDF.

Results & Impact

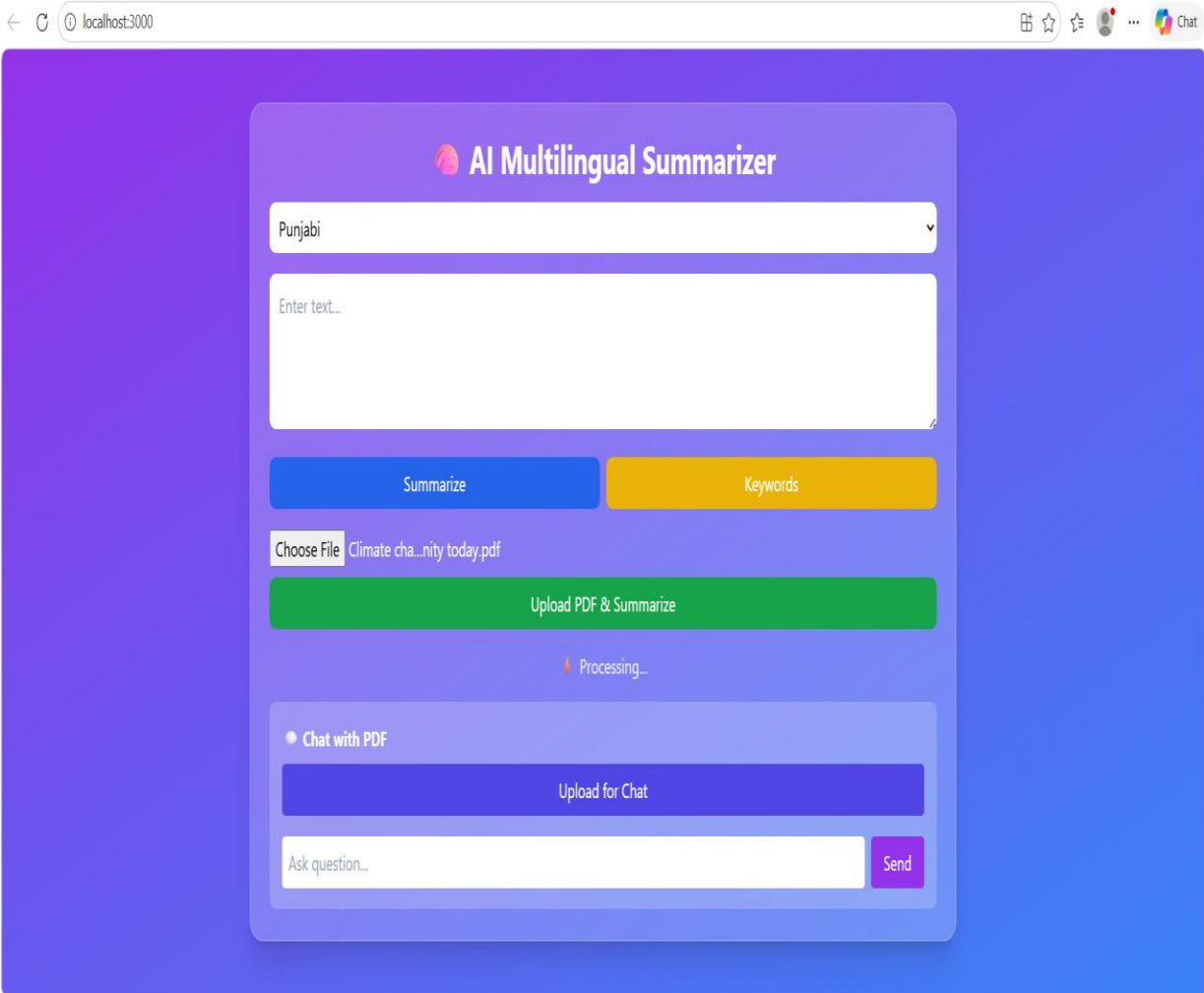
Reduces reading time by ~70%.

Improves productivity for students and professionals.

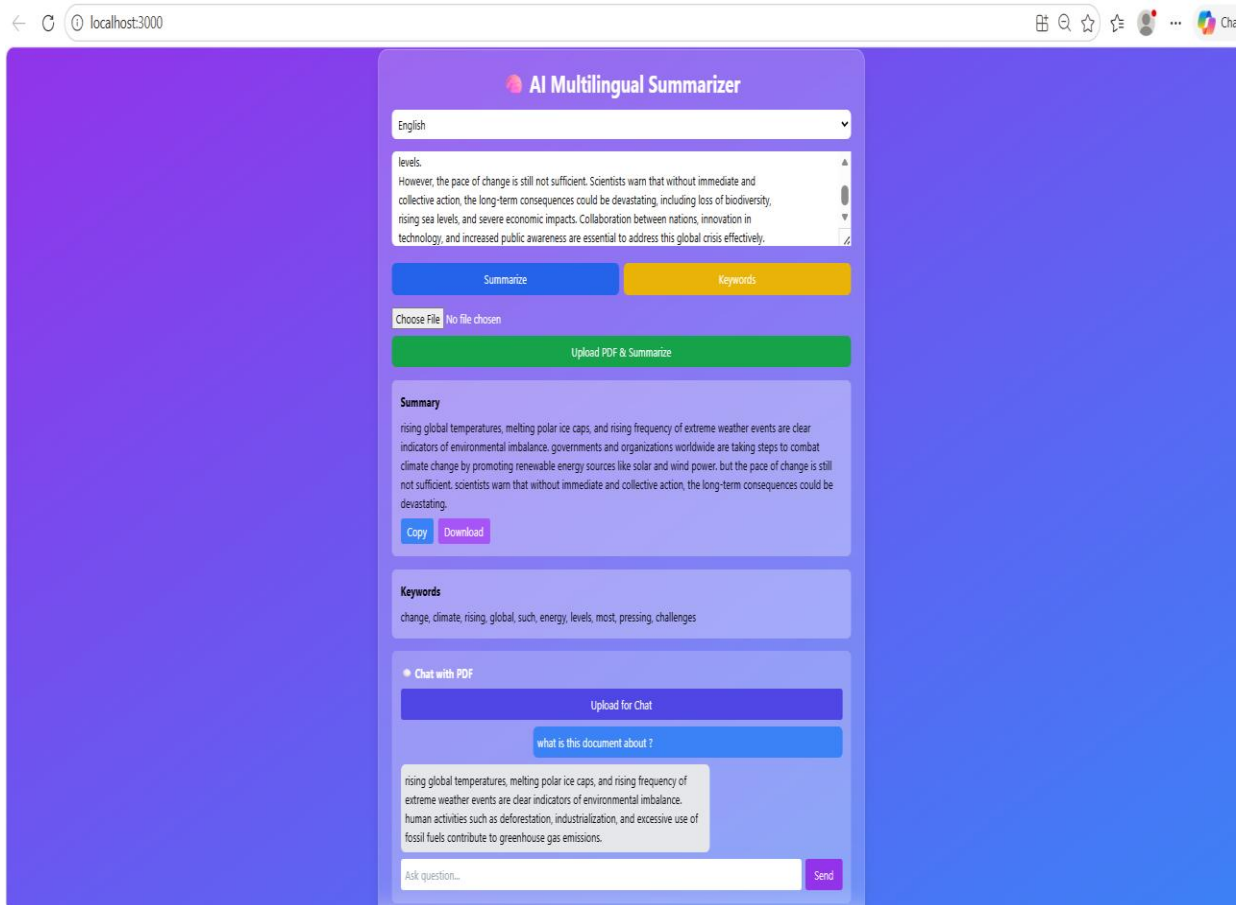
Handles large documents efficiently.

Enhances accessibility via multilingual support.

Dashboard View



Output & Features



Business Value & Recommendations

Can be deployed as SaaS for enterprises.

Useful for legal, healthcare, and education sectors.

Monetization via API subscriptions.

Future scope: larger models, authentication, cloud storage.

GitHub & Conclusion

A scalable AI solution for intelligent summarization.

Combines NLP + full-stack engineering effectively.

Demonstrates strong real-world application potential.

Ready for deployment and portfolio showcase.

View Project on GitHub:- ([Link](#))